MAT313: Differential Geometry for Curves and Surfaces

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Preface: To be written...

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1 Lecture 1

§1.1 Background discussion

Convention. A map is *differentiable* if it has continuous derivatives of all orders at all points of its domain.

A **parametrized differentiable curve** is a differentiable map

$$\alpha : I \to \mathbb{R}^3 \tag{1.1}$$

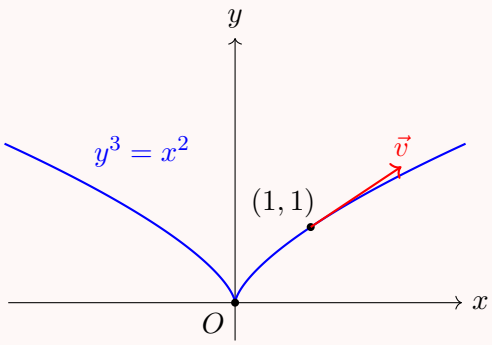
of an open interval[1] $I = (a, b)$ of $\mathbb{R}$ into $\mathbb{R}^3$.

Example 1.1.1

The map

$$\alpha : \mathbb{R} \to \mathbb{R}^3 \quad \text{by} \quad \alpha(t) = (t^3, t^2), \quad \text{where} \quad t \in \mathbb{R} \tag{1.2}$$

is a parametrized differentiable curve. Here, $x = t^3$ and $y = t^2$. Eliminate t to obtain $y^3 = x^2$. Therefore, the curve has the following trace:



In this example, $\alpha'(t) = (3t^2, 2t)$. At $t = t_0 \neq 0$, the slope of the tangent line is $m = \frac{2t_0}{3t_0^2} = \frac{2}{3t_0}$. And the tangent line goes through $\alpha(t_0) = (t_0^3, t_0^2)$. Hence, the equation of the tangent at $t = t_0$ is

$$y - t_0^2 = \frac{2}{3t_0}\left(x - t_0^3\right). \tag{1.3}$$

Definition 1.1.2. Let $\alpha : I \to \mathbb{R}^3$ be a parametrized differentiable curve. For each $t \in I$ where $\alpha'(t) \neq 0$, there is a well-defined straight line which goes through $\alpha(t)$ and contains the vector $\alpha'(t)$. This line is called the **tangent line to** α **at** t. Any point t where $\alpha'(t) = 0$ is a **singular point** of α.

In Example 1.1.1, $t = 0$ is a singular point. We restrict our attention to curves without any singular points.

Definition 1.1.3 (Regular curve)**.** A parametrized differentiable curve $\alpha : I \to \mathbb{R}^3$ is said to be **regular** if $\alpha'(t) \neq 0$ for all $t \in I$.

[1]The word *interval* is taken in a generalised sense, so that we do not exclude the cases $a = -\infty$, $b = \infty$.

Given $t \in I$, the arc length of a regular parametrized curve $\alpha : I \to \mathbb{R}^3$ from the point t_0 is

$$s(t) = \int_{t_0}^{t} |\alpha'(u)| du, \tag{1.4}$$

where

$$|\alpha'(u)| = \sqrt{(x'(u))^2 + (y'(u))^2 + (z'(u))^2} \tag{1.5}$$

given that $\alpha(u) = (x(u), y(u), z(u))$, and so $\alpha'(u) = (x'(u), y'(u), z'(u))$.

The arc length $s(t)$ is a differentiable function of t. Indeed, $|\alpha'| = |\cdot| \circ \alpha'$ is continuous, for being the composition of two continuous functions. Therefore, by the First Fundamental Theorem of Calculus,

$$s(t) = \int_{t_0}^{t} |\alpha'(u)| du \tag{1.6}$$

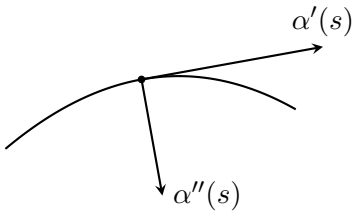
$$\implies s'(t) = |\alpha'(t)|. \tag{1.7}$$

It may well be the case that the parameter t itself is the arch length s, i.e., $s = t$. Then $\frac{ds}{dt} = 1 = |\alpha'(t)|$. And so

$$s = \int_{t_0}^{t} |\alpha'(u)| du = \int_{t_0}^{t} 1 du = t - t_0. \tag{1.8}$$

§1.2 The local theory of curves parametrized by arc length

Let $\alpha : I = (a, b) \to \mathbb{R}^3$ be a curve parametrized by arc length s. We know that the speed is $|\alpha'(s)| = 1$. Hence, the acceleration is normal[2] to the tangent vector at the point. This acceleration vector $\alpha''(s)$ pulls the object and deviates it from the tangent line. If it were zero, the object would have travelled along the tangent line. The more the acceleration, i.e., the more $|\alpha''(s)|$ is, the more the deviation of the trajectory of the object or curve from the tangent line at s.



Definition 1.2.1 (Curvature)**.** Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length $s \in I$. The number

$$\kappa(s) \coloneqq |\alpha''(s)| \tag{1.9}$$

is called the **curvature of** α at s.

Orientation a la do Carmo. The ambient space where the curves and surfaces are living in is $\mathbb{R}^3$. The curve $\alpha(s)$ living in $\mathbb{R}^3$ has a velocity vector

$$\alpha'(s) = \begin{bmatrix} x'(s) \\ y'(x) \\ z'(s) \end{bmatrix}. \tag{1.10}$$

[2] $\alpha(s) \cdot \alpha(s) = 1 \implies 2\alpha''(s) \cdot \alpha'(s) = 0 \implies \alpha''(s) \perp \alpha'(s)$

To discuss the orientation of this curve, do Carmo reverses the direction of traversal as the parameter of the curve increase.

Recall the curvature $\kappa(s)$ of α at s, given by Equation 1.9. At points where $\kappa(s) \neq 0$, a unit vector $n(s)$ in the direction of $\alpha'(s)$ is well-defined by the equation

$$\alpha''(s) = \kappa(s) \cdot n(s). \tag{1.11}$$

From $\alpha'(s) \cdot \alpha'(s) = 1$, one obtains by differentiation $\alpha''(s) \cdot \alpha'(s) = 0$, and from Equation 1.11 one finds that $n(s)$ is normal to $\alpha'(s)$. Indeed,

$$n(s) \cdot \alpha'(s) = \frac{\alpha''(s)}{\kappa(s)} \alpha'(s) = 0. \tag{1.12}$$

$n(s)$ is called the **normal vector at** s.

The plane determined by the unit tangent vector $\alpha'(s)$ and the normal vector $n(s)$ is called the **osculating plane**.

Definition 1.2.2 (Singular points of order 0 and 1)**.** For the arc-parametrized curve $\alpha(s)$, s is a **singular point of order** 0 if $\alpha'(s) = 0$ and it is a **singular point of order** 1 if $\alpha''(s) = 0$ with $\alpha'(s) \neq 0$.

Example 1.2.3 (Planar inflection curve)

The curve $\alpha(t) = (t, t^3, 0)$ is an inflection curve lying in the xy-plane. Note that it's not arc-parametrized. In fact, to understand singular points of order 0 and 1, one doesn't need an arc-parametrized curve necessarily. In this example, $\alpha'(t) = (1, 3t^2, 0)$ reflecting the fact that $\alpha'(t) \neq 0$. It means that the curve has no singular points of order 0. It is a regular curve. One also sees that $\alpha''(t) = (0, 6t, 0)$ so that $\alpha''(0) = (0, 0, 0)$, i.e., 0 is a singular point of order 1.

In what follows, we restrict ourselves to arc-parametrized curve (which is automatically regular) with no singular point of order 1.

We shall denote by

$$t(s) = \alpha'(s) \tag{1.13}$$

the unit tangent vector of α at s so that

$$t'(s) = \alpha''(s) = \kappa(s) n(s) \tag{1.14}$$

The unit vector

$$t(s) \wedge {}^{3} n(s) = b(s). \tag{1.15}$$

is normal to the osculating plane and is called **binormal vector** at s.

$b'(s)$ measures the extent with which the curve is pulled out of the osculating plane. If $b'(s) = 0$, the curve always stays on the osculating plane and is never pulled out off it.

From $b(s) = t(s) \wedge n(s)$, one has

$$b'(s) = t'(s) \wedge n(s) + t(s) \wedge n'(s) = t(s) \wedge n'(s) \tag{1.16}$$

since $t'(s)$ and $n(s)$ are collinear.

In other words, $b'(s) \perp t(s)$. And $b'(s)$ is perpendicular to $b(s)$ itself. And the only way they are satisfied is $b'(s)$ is parallel to $n(s)$, i.e., $b'(s) \parallel n(s)$. And it prompts us to write

$$b'(s) = \tau(s) \cdot n(s) \tag{1.17}$$

for some function $\tau(s)$.

[3] Discussion on wedge/cross product of two vectors in $\mathbb{R}^3$ can be found in [**Car76**, p. 11].

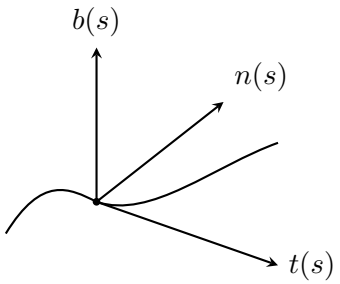


Figure 1.1: The Frenet trihedron at s.

Definition 1.2.4 (Torsion)**.** Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length s such that $\alpha''(s) \neq 0$, $s \in I$. The number $\tau(s)$ defined by $b'(s) = \tau(s)n(s)$ is called the **torsion of** α at s.

Remark 1.2.5. If α is a plane curve (i.e., $\alpha(I)$ is contained in a plane), then the plane of the curve agrees with the osculating plane; hence $\tau \equiv 0$. Conversely, if $\tau \equiv 0$ (and $\kappa \neq 0$) we have $b'(s) = 0$, so that

$$b(s) = b_0 = \text{constant}. \tag{1.18}$$

And therefore,

$$(\alpha(s) \cdot b_0)' = \alpha'(s) \cdot b_0 = 0 \tag{1.19}$$

since $b(s)$ is normal to $\alpha'(s) = t(s)$.

This implies

$$\alpha(s) \cdot b_0 = \text{constant}, \quad \forall s \in I. \tag{1.20}$$

We have $\alpha(s) \cdot b_0 = d =$ some constant for all $s \in I$. For an arbitrary reference point $s_0 \in I$, $[\alpha(s) - \alpha(s_0)] \cdot b_0 = 0$ meaning that the vector joining any two points on the curve $\alpha(s)$ will always be perpendicular to b_0. This means that the curve $\alpha(s)$ lies strictly in a plane perpendicular to b_0.

Note that $\kappa \neq 0$ is essential throughout as

$$n(s) = \frac{\alpha''(s)}{\kappa(s)} \tag{1.21}$$

if $\kappa(s) = 0$ anywhere $n(s)$ ceases to exist and since $b(s) = t(s) \wedge n(s)$, the existence of the constant vector b_0 collapses.

Summary. To each value of the parameter s, we have associated three orthogonal unit vectors $t(s), n(s), b(s)$. The trihedron thus formed is referred to as the **Frenet trihedron** at s. The derivatives $t'(s) = \kappa(s)n(s)$, $b'(s) = \tau(s)n(s)$ of the vectors $t(s)$ and $b(s)$, when expressed in the basis $\{t, n, b\}$, yield geometric entities curvature κ and torsion τ which give us information about the behaviour of α in the neighbourhood of s.

Let us now compute $n'(s)$ using $n(s) = b(s) \wedge t(s)$.

$$\begin{aligned} n'(s) &= b'(s) \wedge t(s) + b(s) \wedge t'(s) && (1.22)\\ &= \tau(s)(n(s) \wedge t(s)) + \kappa(s)(b(s) \wedge n(s)) && (1.23)\\ &= \tau(s)(-b(s)) + \kappa(s)(-t(s)) && (1.24)\\ &= -\tau(s)b(s) - \kappa(s)t(s). && (1.25) \end{aligned}$$

Here, we have used $b'(s) = \tau(s)n(s)$ and $t'(s) = \kappa(s)n(s)$.

We thus have **Frenet formulae**

$$\boxed{\begin{aligned} t'(s) &= \kappa(s)n(s) \\ n'(s) &= -\tau(s)b(s) - \kappa(s)t(s) \\ b'(s) &= \tau(s)n(s) \end{aligned}} \tag{1.26}$$

The tb plane is called **rectifying plane** and the nb plane is called the **normal plane**. The line that contains $n(s)$ and passes through $\alpha(s)$ is called the **principal normal**. And the line that contains $b(s)$ and passes through $\alpha(s)$ is called the **binormal**.

The inverse $R = \frac{1}{\kappa}$ of the curvature is called the **radius of curvature** at s.

Now, let us get back to the orientation discussion of the curve $\alpha(s)$. As $-s$ increases the original parameter s decreases. Define

$$\beta(-s) = \alpha(s). \tag{1.27}$$

so that

$$\beta'(-s) = \frac{d\beta}{d(-s)}(-s) = \frac{d\alpha(s)}{d(-s)} = -\frac{d\alpha(s)}{ds} = -\alpha'(s). \tag{1.28}$$

And

$$\beta''(-s) = \frac{d\beta'(-s)}{d(-s)} = \frac{d}{d(-s)}(-\alpha'(s)) \tag{1.29}$$

$$= \frac{d}{ds}(\alpha'(s)) = \alpha''(s). \tag{1.30}$$

Therefore,

$$\beta''(-s) = \alpha''(s). \tag{1.31}$$

Therefore, $\alpha'(s)$ changes sign if the orientation of the curve reverses while $\alpha''(s)$ and hence curvature remains invariant under orientation reversal of the curve.

Theorem 1.2.6 (Fundamental theorem of the local theory of curves)
Given differentiable functions $\kappa(s) > 0$ (recall $\kappa(s) = |\alpha''(s)|$ so that $\kappa(s) \geq 0$ and $\kappa(s) = 0$ is not allowed as $n(s)$ is defined through $n(s) = \frac{\alpha''(s)}{\kappa(s)}$) and $\tau(s)$, $s \in I$, there exists a regular parametrized curve $\alpha : I \to \mathbb{R}^3$ such that s is the arc length, $\kappa(s)$ is the curvature, and $\tau(s)$ is the torsion of α. (The regular parametrized curve here is actually arc length parametrized which is automatically regular—the existence of such an arc length parametrized curve is claimed in the fundamental theorem). Moreover, any other curve $\widetilde{\alpha}$ satisfying the same conditions, differs from α by a rigid motion; that is there exists an orthogonal linear map ρ of $\mathbb{R}^3$, with positive determinant, and a vector c such that $\widetilde{\alpha} = \rho \circ \alpha + c$.

In the next class, we shall prove the fundamental theorem of local theory of curves.

References

[**Car76**] Manfredo P. do Carmo. *Differential Geometry of Curves and Surfaces.* Englewood Cliffs, NJ: Prentice-Hall, 1976. ISBN: 0-13-212589-7 (cited p. 6)